

Wine Quality Dataset Analysis

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1 Introduction

The data used in this project are laboratory measurements and quality scores for Portuguese Vinho Verde wines from the Minho region in northwestern Portugal. Each record corresponds to one wine sample, described by 11 continuous physicochemical properties measured through routine analytical tests. Alongside these objective inputs, the dataset includes a sensory quality rating on a 0–10 scale as the outcome of interest. The combined file contains 6,497 samples in total, covering both red and white wines. The data are publicly released via the UCI Machine Learning Repository [1], and they are designed to focus on measurable chemical properties rather than external factors like brand, price, or grape variety[2].

What makes this dataset especially appealing for multivariate analysis is that the chemistry is not a collection of isolated measurements. Many of these variables tend to move together because they are shaped by the same underlying processes during fermentation and stabilization. As a result, the 11 recorded properties likely trace a smaller set of latent “profiles” that summarize how a wine is built, such as an acidity related structure, sulfur related preservation, and the alcohol density balance.

In this project, I will address three related questions:

- Compare red versus white wines multivariately by examining whether they occupy distinct regions in a low dimensional PCA representation.
- Summarize dependence among the physicochemical variables through correlation structure, and relate the strongest patterns to quality where relevant.
- Use PCA to identify the dominant modes of variation and interpret the leading components via loading patterns, with attention to how quality varies across the low dimensional space.

2 Data Overview

Overall, the dataset contains 6,497 Portuguese Vinho Verde wines from the Minho region (1,599 red and 4,898 white) and each sample is described by 11 physicochemical variables (including fixed acidity, volatile acidity, citric acidity, residual sugar, chlorides, free sulfur dioxide, total sulfur dioxide, density, pH, sulphates, and alcohol). The raw data include 1,177 duplicate rows. After removing these duplicates, the final analysis dataset contains 5,320 observations. Because the dataset does not provide unique identifiers (e.g., wine, batch, or measurement replicate IDs), these duplicates cannot be interpreted as informative repeated measurements and do not contribute new

information. I therefore remove duplicated rows to prevent repeated chemical profiles from being overweighted. This preprocessing step has negligible impact on the overall distributional patterns and the main conclusions of the analysis, since duplicates replicate existing observations rather than introducing new variability.

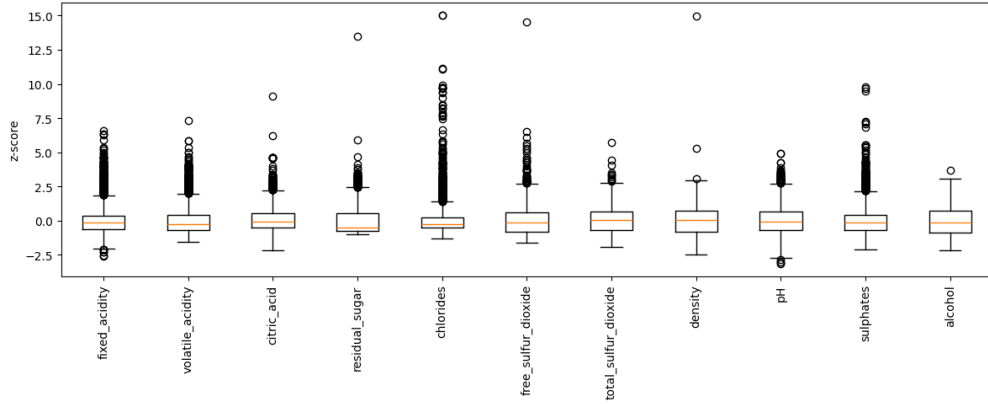
To ground the analysis, I begin with a brief descriptive summary of the dataset. Figure 1(a) reports summary statistics for the 11 physicochemical predictors, providing an initial view of their typical ranges and variability and revealing substantial differences in measurement scale across variables. For example, density varies within a very narrow band (roughly 0.987 to 1.039) with extremely small relative variability, whereas free and total sulfur dioxide span much wider ranges and show much larger dispersion. Several variables also exhibit strong right tails: residual sugar has a median of 2.7 and upper quartile of 7.5 but a maximum of 65.8, and chlorides and both sulfur dioxide measures have maxima that far exceed their upper quartiles, consistent with heavy tailed behavior and occasional extreme values. In contrast, pH and alcohol display comparatively small relative variability. Because a few high variance variables and differing units can dominate distance based analyses, I standardize predictors before PCA and related multivariate methods; for strongly right tailed variables, a log transform (e.g., $\log(1 + \text{residual sugar})$) can be considered for sensitivity analysis.

In addition, the standardized boxplots in Figure 1(b) make it easier to compare relative dispersion across predictors on a common scale. Most variables are centered close to zero after standardization, but several exhibit pronounced right tails and many high-end outliers, most notably the sulfur dioxide measures, chlorides, and residual sugar. By contrast, pH and alcohol appear comparatively tight with fewer extreme standardized values. This figure therefore complements the raw-unit summary by highlighting which predictors contribute the most atypical observations once scale differences are removed.

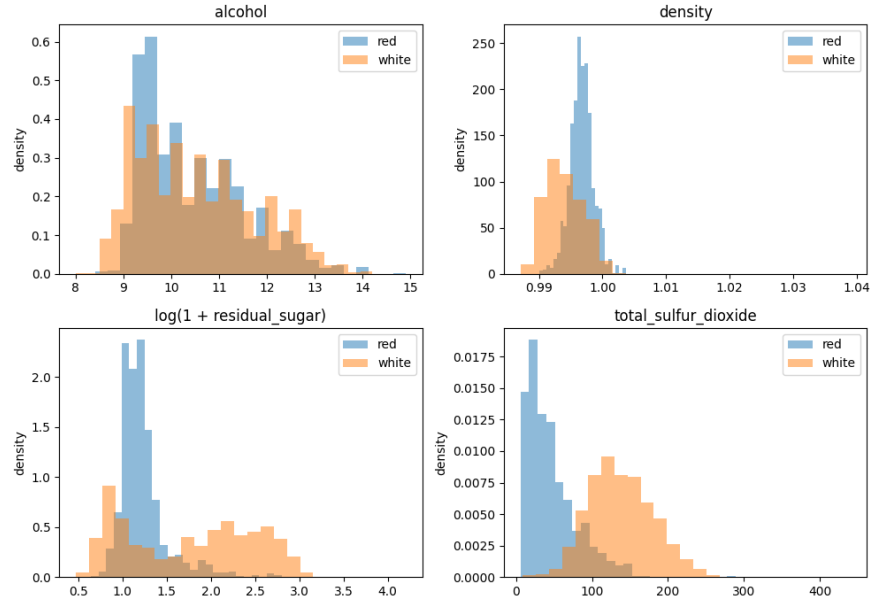
From these univariate plots (Figure 1(c)), the most visible patterns are type shifts rather than just spread. To keep the overview concise, I focus on four representative variables that are easy to interpret and show clear contrasts between red and white wines. This selection also anticipates the main multivariate gradients later revealed by PCA, where alcohol and residual sugar drive a central tradeoff while density and sulphur dioxide align with complementary structure across the samples. Density shows a clear horizontal shift, with red wines concentrated at slightly higher densities and white wines shifted lower, suggesting systematic differences in the physical profile of the two groups. Residual sugar (after a $\log_{1p}$ transform to reduce right skew and compress extreme values) separates even more strongly: white wines have a broader distribution with substantial mass at higher values, while reds cluster tightly at lower sugar levels. Total sulfur dioxide also differs markedly by type, with white wines centered at much higher concentrations and red wines mostly at lower levels. Alcohol largely overlaps by type, but red wines are more concentrated (a sharper peak) while white wines are more dispersed (a flatter peak), with no major shift in the center.

Variable	mean	std	min	25%	50%	75%	max	IQR	range	CV
residual_sugar	5.0485	4.5002	0.6000	1.8000	2.7000	7.5000	65.8000	5.7000	65.2000	0.8914
chlorides	0.0567	0.0369	0.0090	0.0380	0.0470	0.0660	0.6110	0.0280	0.6020	0.6503
free_sulfur_dioxide	30.0367	17.8050	1.0000	16.0000	28.0000	41.0000	289.0000	25.0000	288.0000	0.5928
total_sulfur_dioxide	114.1090	56.7742	6.0000	74.0000	116.0000	153.2500	440.0000	79.2500	434.0000	0.4975
volatile_acidity	0.3441	0.1682	0.0800	0.2300	0.3000	0.4100	1.5800	0.1800	1.5000	0.4889
citric_acid	0.3185	0.1472	0.0000	0.2400	0.3100	0.4000	1.6600	0.1600	1.6600	0.4620
sulphates	0.5334	0.1497	0.2200	0.4300	0.5100	0.6000	2.0000	0.1700	1.7800	0.2808
fixed_acidity	7.2152	1.3197	3.8000	6.4000	7.0000	7.7000	15.9000	1.3000	12.1000	0.1829
alcohol	10.5492	1.1859	8.0000	9.5000	10.4000	11.4000	14.9000	1.9000	6.9000	0.1124
pH	3.2247	0.1604	2.7200	3.1100	3.2100	3.3300	4.0100	0.2200	1.2900	0.0497
density	0.9945	0.0030	0.9871	0.9922	0.9947	0.9968	1.0390	0.0046	0.0519	0.0030

(a) Descriptive statistics for the 11 predictors, showing large differences in scale.



(b) Standardized boxplots (z scores) comparing dispersion and outliers across predictors.



(c) Distributions of four predictors by wine type (alcohol, density, $\log(1 + \text{residual sugar})$, total sulfur dioxide), showing strong type separation for sugar and sulfur measures.

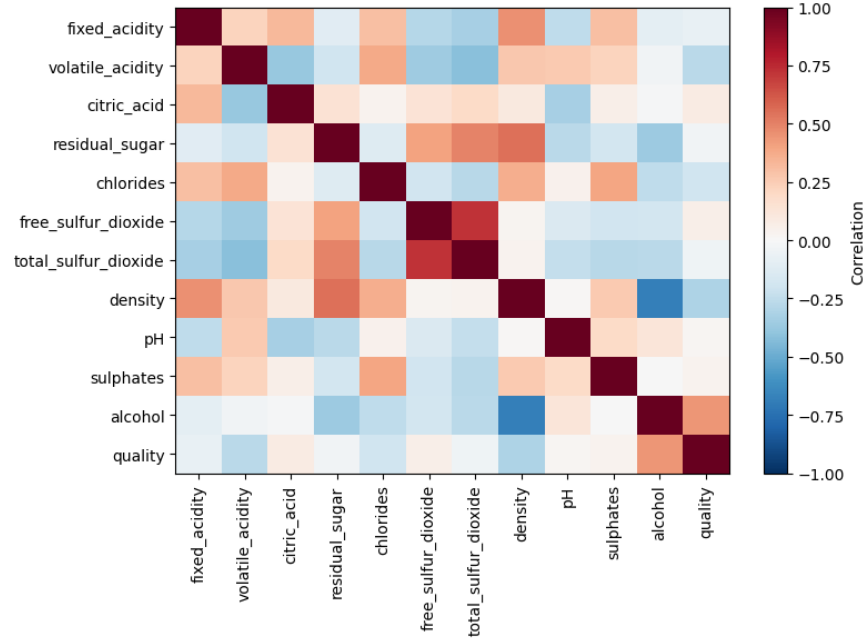
Figure 1: Data overview of the wine dataset. Panel (a) summarizes the 11 physicochemical predictors and their scales. Panel (b) shows standardized boxplots for comparing dispersion and outliers across variables. Panel (c) compares four representative predictors by wine type (alcohol, density, $\log(1 + \text{residual sugar})$, total sulfur dioxide), highlighting key red white differences that motivate the multivariate analysis.

3 Correlation Structure

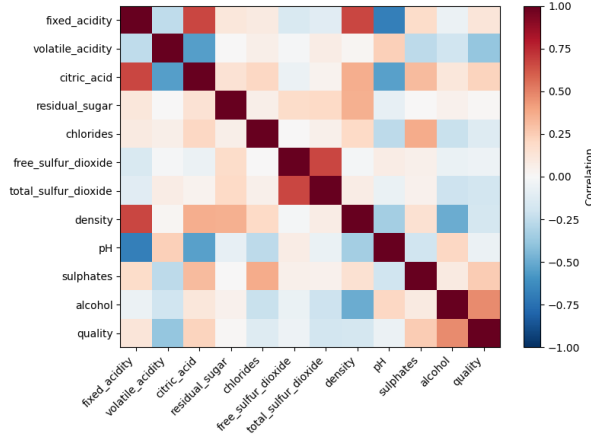
In this section, I characterize the correlation structure overall and within each wine type to anticipate the directions that dominate the PCA score space. The correlation matrices show several consistent relationships among the physicochemical predictors, and including ‘quality’ helps connect these patterns to the response. In the overall matrix, the strongest positive correlation is between free sulfur dioxide and total sulfur dioxide ($r = 0.721$), which is expected since these measures quantify related aspects of sulfite content. A second clear block involves sweetness and density: residual sugar is positively correlated with density ($r = 0.553$) and with total sulfur dioxide ($r = 0.495$), suggesting that sweeter wines tend to be denser and also tend to have higher sulfur dioxide levels. The most prominent negative relationship in the overall data is between alcohol and density ($r = -0.687$), reflecting that higher alcohol wines are typically less dense. Several acidity variables also move together, for example fixed acidity is positively correlated with citric acid ($r = 0.324$) and negatively with pH ($r = -0.253$), consistent with pH acting as an inverse measure of acidity.

When stratifying by wine type, some associations strengthen, indicating that mixing red and white can partially mask within type structure. For red wines, the acidity block becomes much clearer: fixed acidity is strongly positively correlated with citric acid ($r = 0.672$) and strongly negatively correlated with pH ($r = -0.683$), while citric acid is also negatively correlated with volatile acidity ($r = -0.552$). For white wines, the sugar density relationship is substantially stronger than in the pooled data: residual sugar is very strongly correlated with density ($r = 0.839$) and is moderately correlated with total sulfur dioxide ($r = 0.401$). The alcohol density trade off is strong in both types, but it is especially pronounced in white wines (red $r = -0.496$, white $r = -0.780$). These differences are visible in the heatmaps as stronger colour contrasts within each type, highlighting that red and white wines have different ranges and dependence patterns across the same set of measurements.

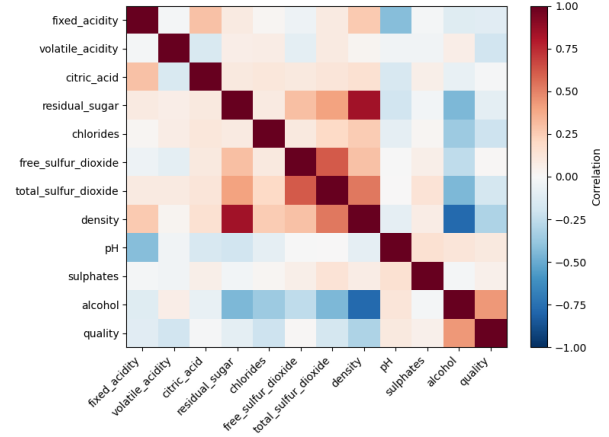
Including ‘quality’ shows that quality is not strongly tied to a single variable, but there are consistent directional signals. In the overall matrix, quality is most positively correlated with alcohol ($r = 0.444$) and most negatively correlated with density ($r = -0.306$) and volatile acidity ($r = -0.266$). This pattern largely holds within types: for red wines, quality has its strongest positive association with alcohol ($r = 0.476$) and strongest negative association with volatile acidity ($r = -0.391$), while for white wines quality remains positively associated with alcohol ($r = 0.436$) and negatively associated with density ($r = -0.307$) and chlorides ($r = -0.210$). Overall, these matrices suggest two main dependence structures in the data, a sulfur dioxide block and a sugar density block, plus a strong alcohol density trade off, and quality appears to align most with the alcohol density axis and, particularly for reds, lower volatile acidity. These correlation patterns also help interpret the PCA results. Because PCA is applied to standardized variables, the leading components tend to align with groups of predictors that co-vary strongly. In our data, the tight free/total sulfur dioxide relationship and the residual sugar–density block correspond to a shared direction of variation, while the strong negative alcohol–density association forms another dominant contrast. This is consistent with the PCA results, where the first two components capture these same co-variation structures and provide a low-dimensional summary of how the predictors move together across wines.



(a) Correlation heatmap for the full dataset, summarizing overall linear associations among the predictors (and quality).



(b) Red wine correlations, showing within type structure after removing red white mixing, with a clearer alcohol–density tradeoff and weaker links between sugar and sulphur measures.



(c) White wine correlations, showing within type structure, where residual sugar aligns more strongly with density and sulphur measures and alcohol tends to oppose density.

Figure 2: Correlation heatmaps for the full dataset and separately for red and white wines. The full sample (a) reveals structured correlation blocks, most notably the free total sulphur dioxide pair and the sugar density cluster, while the stratified panels (b)–(c) isolate within type relationships and help distinguish genuine chemical associations from correlations induced by red white differences.

4 Principal Component Analysis

I performed PCA on the standardized physicochemical predictors, with each variable centered and scaled to unit variance so that the components summarize correlation structure rather than being dominated by measurement scale. Mardia’s multivariate normality test strongly rejects joint normality for the $p = 11$ predictors in the cleaned dataset ($n = 5320$). The multivariate skewness statistic is $b_{1,p} = 109.02$, corresponding to $\chi^2 = 96664.55$ with 286 degrees of freedom ($p \approx 0$). The multivariate kurtosis statistic is $b_{2,p} = 445.07$ with standardized $z = 651.40$ ($p \approx 0$). These results are consistent with the observed skewness and extreme values in several predictors, so the PCA results are interpreted as dominant linear patterns in the sample and may be somewhat sensitive to extreme observations.

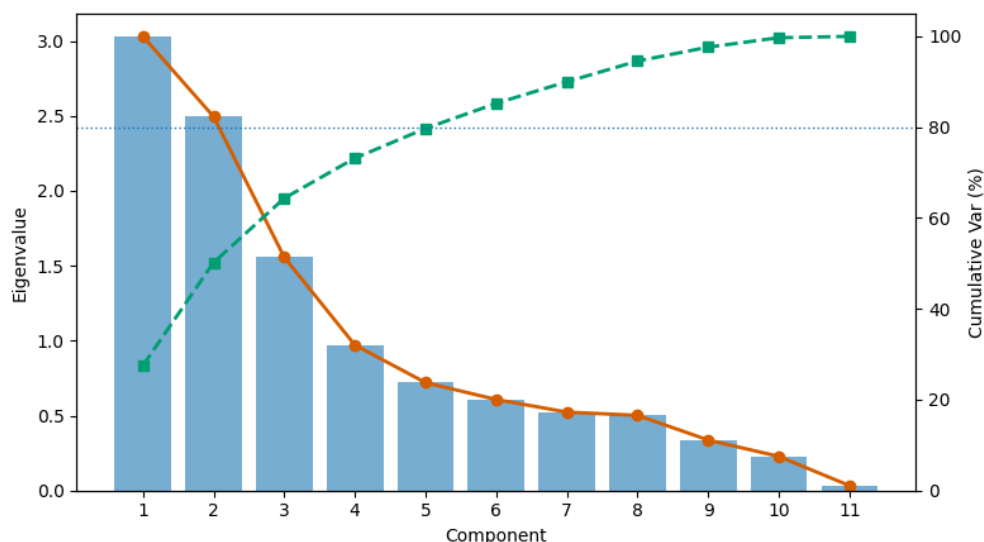
Figure 3(a) suggests that most systematic variation is concentrated in the leading components, with a clear elbow after the first few PCs. Using the Kaiser criterion (eigenvalue greater than 1) supports retaining PC1 through PC3, since the first three components exceed 1 while later components contribute relatively small additional variance. In particular, PC1 and PC2 explain 27.2% and 22.5% of the standardized variance, respectively, capturing about half of the total variation while remaining easy to visualize. Based on the dominant loadings, PC1 is labelled the *Production Style & Preservation Axis* because it contrasts wines with higher residual sugar and sulfur dioxide (positive direction) against those with higher chlorides (negative direction), while PC2 is labelled the *Fermentation & Body Axis* because it is driven primarily by an alcohol (negative) versus density (positive) trade off. Although a three component solution is reasonable by the Kaiser rule, PC3 mainly increases variance explained rather than clarifying the dominant structure emphasized in the score plot and loading patterns, so the interpretation below focuses on PC1 and PC2.

The PC1 and PC2 score plot in Figure 3(b) shows a strong separation between red and white wines driven primarily by PC1 (*Production Style & Preservation Axis*). Red wines concentrate on the negative side of PC1, while white wines lie mostly on the positive side, with limited overlap near the origin. The loading directions indicate that positive PC1 is aligned with higher residual sugar and higher sulfur dioxide measures, while negative PC1 is aligned with higher chlorides. This suggests that the red white separation reflects systematic differences in these chemical profiles, with white wines tending toward higher residual sugar and sulfur dioxide and red wines tending toward higher chloride levels in the standardized space.

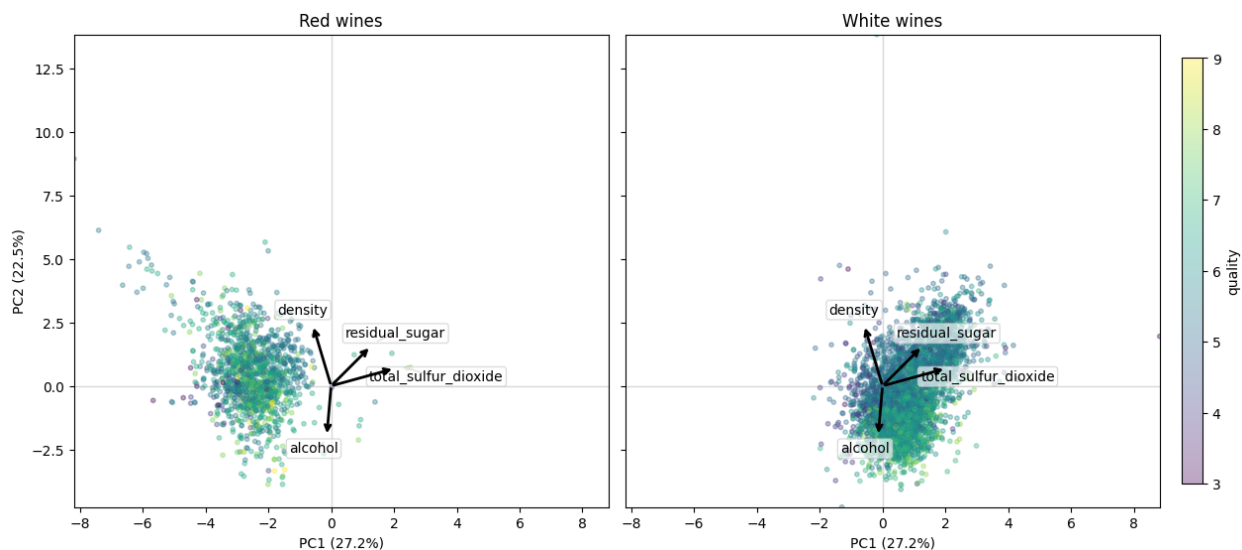
PC2 (*Fermentation & Body Axis*) captures an additional within type gradient that does not primarily separate reds from whites. The dominant contrast on PC2 is density versus alcohol: observations with higher PC2 scores tend to have higher density, while observations with lower PC2 scores tend to have higher alcohol. This axis is also relevant for quality variation. Within each wine type, higher quality observations appear more frequently toward the lower PC2 region, consistent with wines that have relatively higher alcohol and lower density, whereas lower quality observations are more common toward the upper PC2 region. Although this pattern is not perfectly monotone, it indicates that quality differences are more closely associated with within type variation along the density alcohol contrast than with the across type separation on PC1.

The loading plot in Figure 3(c) provides a numerical summary supporting the biplot interpretation by showing which variables contribute most to PC1 and PC2 and in which direction. For PC1, residual sugar and the sulfur dioxide variables have strong positive loadings, while variables associated with the negative side of PC1 point in the opposite direction, consistent with the red white

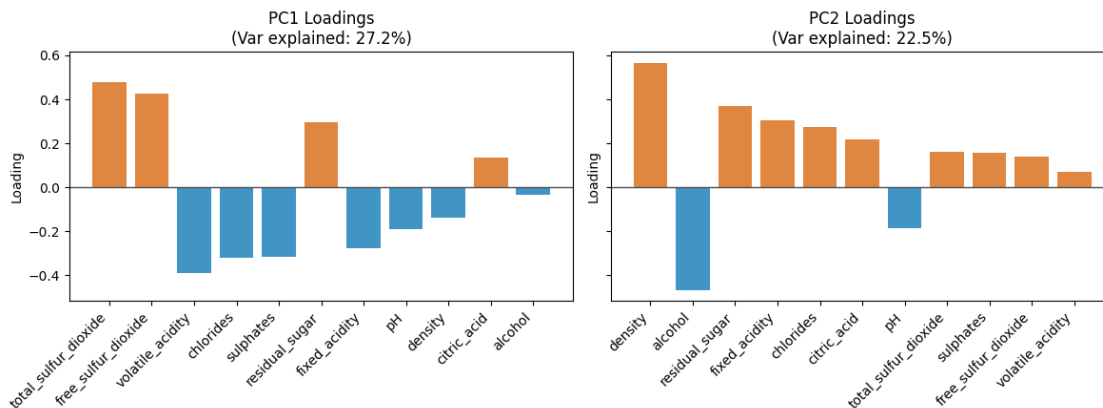
separation in the score plot. For PC2, density loads strongly positive and alcohol loads strongly negative, reinforcing the interpretation of PC2 as a density alcohol trade off that varies within both red and white wines. The sign of a loading indicates the direction of association with a component, and because PCA directions are identifiable only up to an overall sign flip, the interpretation relies on relative signs and magnitudes rather than the absolute choice of positive versus negative.



(a) Scree plot of PCA eigenvalues with cumulative variance explained, used to assess how many components are needed to summarize the standardized predictors.



(b) PC1–PC2 score plots shown separately for red and white wines, with points coloured by quality; PC1 (Production Style & Preservation Axis) and PC2 (Fermentation & Body Axis) are annotated with arrows indicating the strongest loading directions for key physicochemical variables.



(c) Bar plots of variable loadings for PC1 and PC2, ordered by absolute loading magnitude; bar sign indicates the direction of association with each component.

Figure 3: PCA summary of the wine dataset. Panel (a) shows the elbow at PC3, justifying a low-dimensional representation. Panel (b) illustrates the red-white separation along the Production Style Axis (PC1) and quality variation along the Fermentation Axis (PC2). Panel (c) reports PC1 and PC2 loadings, highlighting the variables that drive the dominant multivariate contrasts observed in (b) and supporting interpretation of the component directions.

5 Discussion

Overall, the EDA and PCA indicate that wine type is a major organizing factor in the physicochemical measurements. In the PC1–PC2 space, the dominant separation is along PC1, which we label the *Production Style & Preservation Axis* because it contrasts residual sugar and sulfur dioxide measures against chlorides. This aligns with practical differences in how red and white wines are produced and stabilized, so it is unsurprising that the chemistry clusters by production style even before considering quality.

At the same time, the results also show substantial variation within each wine type. This within type variability likely reflects differences in grape variety, fermentation conditions, and winemaker choices, so it is expected that not every red or every white wine has the same chemical profile. Much of this within type structure is captured by PC2, labelled the *Fermentation & Body Axis*, which reflects a density versus alcohol contrast and appears to be where quality varies most clearly within both red and white wines. The formal rejection of multivariate normality by Mardia’s test further suggests that the data is heavily influenced by skewness and potential outliers, implying that while PCA effectively summarizes the variance, the resulting components may be sensitive to extreme chemical profiles rather than just the central bulk of the data. Because these analyses are descriptive and PCA is variance driven, the patterns should be interpreted as a summary of how the measurements vary together rather than evidence of causality.

To strengthen the conclusions, I reran the same correlation and PCA analyses on the dataset with duplicated rows retained, and the main patterns remained qualitatively unchanged. Remaining robustness checks could include assessing sensitivity to influential outliers and comparing PCA results to a robust alternative, as well as repeating PCA within red and within white subsets to confirm that the Fermentation & Body Axis and its association with quality are consistent within each type rather than being an artifact of the overall type separation.

References

- [1] UCI Machine Learning Repository. *Wine Quality Data Set* (ID 186). DOI: 10.24432/C56S3T.
- [2] P. Cortez, A. Cerdeira, F. Almeida, T. Matos, and J. Reis. Modeling wine preferences by data mining from physicochemical properties. *Decision Support Systems*, 47(4):547–553, 2009. DOI: 10.1016/j.dss.2009.05.016.